2024 International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT) | 979-8-3503-8427-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/TQCEBT59414.2024.10545084

Decoding Customer Lifetime Value to Unlock Business Success with Predictive Machine Learning Approach

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***Abstract*:-This study highlights how crucial customers are for a company's success who directly impacts revenue and overall business value. This study focuses on analysis of customer lifetime value, the research uses data from 5000 customers with 8 important features with the main goal of predicting customer lifetime value. Business leaders often face choices about where to invest in marketing, like loyalty programs, incentives and ads or nothing. The study suggests that customer lifetime value is a key metric for making smart decisions, which measures how much a customer spends over their time with a company. To predict this value, the research explored different machine learning models – linear regression, decision tree regressor, random forest, and AutoML regressor. Each model is checked for how well it predicts customer spending habits. The results show that AutoML regression stands out for its accuracy without overcomplicating things. This study offers insights for businesses looking to improve their customer-focused strategies and long-term success.**

Keywords:- Customer lifetime value prediction, decision tree, random forest, linear regression, AutoML regression

I. INTRODUCTION

Customers are critical important part of any organization because in their absence there would be no revenue, profit and company's market value [1]. Business executives are always in the dilemma to invest in marketing campaigns like loyalty program, advertising or none of it[2].In last few years , the customer value for the business has been getting attention among the strategist as a key ingredient to business competitive strategy[3]. Customer lifetime value need to be recognised as key intangible asset of the business and it should not only be a part of business strategies but need to be considered in overall business valuation process [4] .Customer lifetime values basically measures the total money amount to be spent by customer during its lifetime association with business. Companies can effectively take a strategic decision related to the customer segmentation, offering, incentives and their retention by understanding customer lifetime value[5]. Customer lifetime values takes into account for customer's every order, engagement, loyalty and help in evaluation of business health[6].In today's dynamic business setting, organisations are shifting their focus from market to individual customers preferences and place the customer at the core of business strategies[7]. In these given conditions companies need to prioritize customer relationship management to stay business afloat, to improve customer satisfaction and retention [8][9]. The economics of customer lifetime value encourages a paradigm shift, where customers are seen through the lens of continuous relationships rather than any business transactions[10].

This study focuses on customer lifetime value prediction using various machine learning predictive models. The first section of this paper introduces to the concept of customer lifetime value and second part discusses about previous studies for the same. Third part of paper introduces to the features and target used in the dataset along with the predictive machine learning models used in the study. In fourth and fifth part discusses about the model outcome and conclusion, future scope of the study respectively.

II. RELATED WORKS

[5]examined the customer classification by using random forest model for online retail consumers. This study used random search technique to identify the best hyperparameter set and study found that model performance with random search trained model improved by 2% and 6% compared to the feature selection trained model and Adaboost trained model respectively.[7] used lightGBM model to classify 3,246 customers lifetime value of ride-hailing company where the data is collected from the January 2018 to September 2018. Researcher's used grid search and 5-fold method to identify model's best parameter to train the model. This study results shows that the GOSS and DART models are superior compared to other models.[9] in their study applied k-means clustering model to the data collected from a distributor of pharmaceutical and medical device products for 7486 transactions data between 1/1/2019 to 31/12/2019. This study found 8 resulting clustering and model indicates frequency, monetary benefit, recency and length are key features.[11] analysed customer lifetime value with two stage neural network with the data from insurance company. Non-relationship of user's preferences effectively captured by the proposed model and have an edge over linear model.[6] built a model to predict customer lifetime value of 9,134 different bank customers with 27 different attributes. This study used four models to predict customer lifetime value i.e. linear

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regression, decision tree regression, random forest regression and auto ml regression. The results of these models for R2 are 26%, 89.8%, 89.9% and 90.10% respectively. Out of these models AutoML regression model found as best fit. [12] in their study analysed customer lifetime value prediction with the XGBoost and k-means clustering and study predicted successfully customer lifetime value, loyalty with 77-78% accuracy for upcoming years.[13] examined the customer lifetime value prediction with IBM Watson dataset and study findings suggests that random forest and decision tree models are performing best consistently compared to the linear regression and support vector machine models.

III. PROPOSED WORK

This study is focused towards the customer lifetime value prediction. In this study 7 features are used to predict customer lifetime value with 5000 dataset. The customer lifetime value is predicted with linear regression, decision tree regressor, random forest regressor and AutoML regressor.

A. Linear regression model

In first step, linear regression machine learning is trained and tested on the 7 features to predict customer life time value. In second step, features with variance inflation factor above 10 are removed to avoid the problem of multi-collinearity. The predictive ability of model is the validated with R^2 value, root mean square error (RMSE) and mean absolute error values.

B. Decision tree regressor

Decision tree regression algorithm is powerful tool used for prediction of customer lifetime value. This algorithminitially analyses the entire dataset, begins with the root node. Then data starts to split at every decision node and this process goes on subsequently. The prediction is a final decision that remains with the leaf node. The problem with the decision tree model is that it is prone to the model overfitting.

In the first step, decision tree regressor model is trained and tested and then model results are validated. In second step, if needed to improve the model performance hyperparameter tuning is used. Hyperparameter tuning in a decision tree regressor involves adjustment with decision tree settings before training process. These adjustments relate to the settings of depth of tree, minimum numbers of split, minimum samples leaf, selection of maximum features, tree pruning.

C. Random forest

Random forest is powerful machine learning algorithm for data classification and prediction. This technique is based on idea of ensemble model.The underlying idea of random forest is to create a multiple decision tree, each tree is independently trained on random subset of the features for prediction.

In step 1, model is trained and tested with random forest algorithm. In step 2, hyperparameter tuning is used to improve the model performance if needed. Hyperparameters tuning for random forest can be number of estimators, maximum depth, minimum samples per split, maximum features in splitting in a node etc.

D. AutoML regression

AutoML (Automated Machine Learning) is an innovative approach to handle the data for prediction.Traditional regression modelling where expertise is required to handle feature engineering, model selection and algorithm selection. AutoML models automatically select the most suitable regression algorithm to given dataset.

Table I presents the definition of features and target variable used in the dataset.

TABLE I. VARIABLE MODEL DEFINITION

Variable name	Definition
Purchase Frequency	Number of purchases made by the customer
Average Purchase Value	Average amount spent by the customer in each transaction
Customer Satisfaction Score	Rating indicating the level of satisfaction
Customer Acquisition Cost	Cost associated with acquiring a new customer
Retention Rate	Percentage of customers retained over a specific period
Time Since Last Purchase Months	Duration since the customer's last transaction
Age	Age of the customer
Churn Rate	Percentage of customers lost over a given period
Customer Lifetime Value	Total revenue from a customer throughout their entire relationship

IV. MODEL EXPERIMENTATION

This section presents the customer lifetime value prediction outcome with linear regression, decision tree regressor, random forest regressor and AutoML regressor.

TABLE II. LINEAR REGRESSION RESULTS

Model	R^2	RMSE	MAE
Train	0.740	1052.15	674.86
Test	0.745	1037.00	694.00
	VIF Values		
Purchase Frequency		3.96	
Customer Acquisition Cost		7.41	
Time Since Last Purchase Months		4.29	
Age		7.54	
Churn Rate		8.96	

Table II presents the linear regression model outcome, this model is trained on the 7 features to predict customer lifetime value. Initially the model was ran with all the features and features having variance inflation factor vales above 10 are excluded. These variable incudes retention rate, average purchase value and customer satisfaction score. The findings suggests that final regression model includes 5 features however the model's customer lifetime value predictive ability is weak as R^2 value is 74% and 74.50 % for train and test model respectively.

A. Decision tree regressor

TABLE III. DECISION TREE BASIC MODEL

Model	R^2	RMSE	MAE
Train	1	0.00	0.00
Test	0.95	458.32	270.87

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Decision tree basic model results are presented in table III. The results shows that the R^2 value for training model is 1 which suggests the model correctly recognises customer lifetime value. However on the test, model can correctly recognises 95% of data. Hence it is confirmed that the model is performing well on training data but not on test data which indicates overfitting of this model.To address this overfitting problem of model hyperparameter tuning is used.Root mean square error (RMSE) value is 0.00 and mean absolute error (MAE) 0.00 on trained model indicatesoverfitting problem.

TABLE IV. HYPERPARAMETER TUNING

Model	**R^2**	**RMSE**	**MAE**
Train	0.98	217.24	142.02
Test	0.95	446.69	278.16
Hypermeters specification			
Maximum depth		10	
Min. samples leaf		1	
Minimum samples split		10	

Table IV presents the outcome of decision tree hyperparameter tuning results. Maximum depth of 10 indicates maximum depth of decision of tree is 10 levels i.e. this is length of longest path from the root node to a leaf node. Minimum sample leaf is kept at 1 to become more complex and detailed decision tree. Minimum samples split 10 suggests to get split in internal code minimum 10 samples are required, this controls the growth of tree. The findings suggests that test R^2 value remains constant in basic decision tree and hyperparameter tuning regression model at 95%. However, problem of overfitting is resolved as RMSE and MAE values are 217.24 and 142.02 for training modeland in previous model these values are 0.00 for both.

TABLE V. RANDOM FOREST REGRESSOR WITH HYPERPARAMETER TUNING

Model	**R^2**	**RMSE**	**MAE**
Train	0.99	146.75	82.19
Test	0.96	362.29	212.91
Best Hyperparameters specification			
Maximum depth		None	
Maximum features		Log2	
Minimum samples leaf		1	
Minimum samples split		2	
N estimators		200	

Table V random forest regressor results and findings indicates that R^2 value for training and testing model are 99% and 96% respectively. This suggests model performs very well on both the training and testing model. Root-mean square error and mean absolute error values are reasonably low, suggesting model accuracy. This outcome is based on the hyperparameter tuning. In hyperparameter settings maximum depth none suggests that tree is allowed to grow and expand without any constraints, then leaf node should contain at least 1 sample, minimum numbers required to split further is 2 samples, random forest is made up with 200 trees and model considers log2 of features in the dataset.

B. AutoML regression

TABLE VI. AUTOML REGRESSION

Model	**R^2**	**RMSE**	**MAE**
Train	0.9988	70.97	43.62
Test	0.9934	166.22	87.00
AutoML regression model specification			
XGB Regressor		Ride CV(input matrix)	
Learning rate		0	
Maximum depth		6	
Min child weight		3	
N estimators		100	
N jobs		1	
Sub-sample		0.85	
Verbosity		0	

Table VI represents results for AutoML regression, the study findings are based on XGBregressor. Model performance is very high with R^2 of 99.88% and 99.34% for train and test model respectively and there is no model overfitting problem visible. Root mean square error and mean absolute error values are lowest when compared to other models.

V. CONCLUSION AND FUTURE WORK

In this study various regression models are used to predict the customer lifetime value with 7 features and dataset includes 5000 customers information. The linear regression model is trained and tested by excluding the features with high variance inflation factor.However, the model found with moderate predictive ability. The decision tree in basic form suffered from model overfitting problem with R^2 of 100 % on training and 95 % on testing. This problem was addressed with hyperparameter turning. The random forest regressor with hyperparameter tuning, showcases high predictive accuracy with R^2 values of 99% and 96% for training and testing, respectively. This model's adaptability to data is attributed to parameters such as unlimited maximum depth, 200 trees, and consideration of log2 features.The AutoML regression which employed XGBRegressor with specific hyperparametersfound as the most effective model. With R^2 values of 99.88% for training and 99.34% for testing, the AutoML model showcases remarkable accuracy without overfitting concerns. Lower RMSE and MAE values further affirm its superior predictive performance. Hence this study confirms and recommends to use AutoML regression model to predict customer lifetime value as compared to the linear regression, decision tree and random forest model considering overall model performance.

A. Scope for future work

This study opens various channels for further studies in customer lifetime value prediction. In future researchers can give more focus on actual data to relate theory with the practise and to understand the underlying phenomenon from close. The use of advanced analytical techniques like explainable AI methods, SHAP values, feature engineering, ensemble methods can provide additional information for customer lifetime value. There is a need to evaluate and reconfirm this study results with exclusive and detailed dataset.

REFERENCES

[1] S. Gupta and D. R. Lehman, "Customers as assets," Journal of Interactive Marketing, vol. 17, no. 1, pp. 9-24, 2023.

Authorized licensed use limited to: Cambridge Institute of Technology - Bengaluru. Downloaded on March 12,2026 at 09:24:08 UTC from IEEE Xplore. Restrictions apply.

[2] R. M. Rust, K. Lemon and V. Zeithaml, "Return on marketing: Using customer equity to focus marketing strategy," Journal of Marketing, vol. 68, no. 1, pp. 109-127, 2004.

[3] D. Wayne, J. Kamel and I. Sinha, "Customer Value Analysis in a Heterogeneous Market," Strategic Management Journal, vol. 22, no. 9, pp. 845-857, 2001.

[4] S. Gupta and D. Lehmannand, "Valuing customers," Journal of Marketing Research, vol. 41, no. 1, pp. 7-18, 2004.

[5] T. T. Win and K. S. Bo, "Predicting Customer Class using Customer Lifetime Value with Random Forest Algorithm," in International Conference on Advanced Information Technologies (ICAIT), Yangon, Myanmar,, 2020.

[6] M. Surti, V. Shah, S. Bharati and R. Gupta, "ustomer Lifetime Value Prediction of an Insurance Company using Regression Models," in International Conference for Advancement in Technology (ICONAT), Goa, India, 2023.

[7] J. He and L. Chen, "The Research of Customer Lifetime Value Related to Risk Factors in the Internet Business : Exampled by Online Car-hailing Industry," in IEEE/ACIS 18th International Conference on Computer and Information Science (ICIS), Bejin, China, 2019.

[8] P. P. Pramono, I. Surjandari and L. Enrico, "Estimating Customer Segmentation based on Customer Lifetime Value Using Two-Stage Clustering Method," in 16th International Conference on Service Systems and Service Management (ICSSSM), Shenzhen, China, 2019.

[9] R. Rahmadianti, A. Dhini and E. Laoh, "Estimating Customer Lifetime Value using LRFM Model in Pharmaceutical and Medical Device Distribution Company,," in International Conference on ICT for Smart Society (ICISS), Bandung, Indonesia,, 2020.

[10] P. D. Berger and N. Nasr, "Customer lifetime value: Marketing models and applications," Journal of Interactive Marketing, vol. 12, no. 1, pp. 17-30, 1998.

[11] D. G, D. A, D. J, M. I and G. D, "Maximizing Customer Lifetime Value using Stacked Neural Networks: An Insurance Industry Application," in 8th IEEE International Conference On Machine Learning And Applications (ICMLA) , Boca Raton, FL, USA,, 2019.

[12] M. Marius and S. Berman, "Customer Lifetime Value Prediction with K-means Clustering and XGBoost,," in International Conference on Advances in Social Networks Analysis and Mining (ASONAM), Istanbul, Turkey, 2022.

[13] H. Kailash, K. Kanwar , S. Sonia and R. Kant, "Machine Learning Algorithms for Predicting Customers' Lifetime Value: A Systematic Evaluation," in 3rd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE), Greater Noida, India, 2023.

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